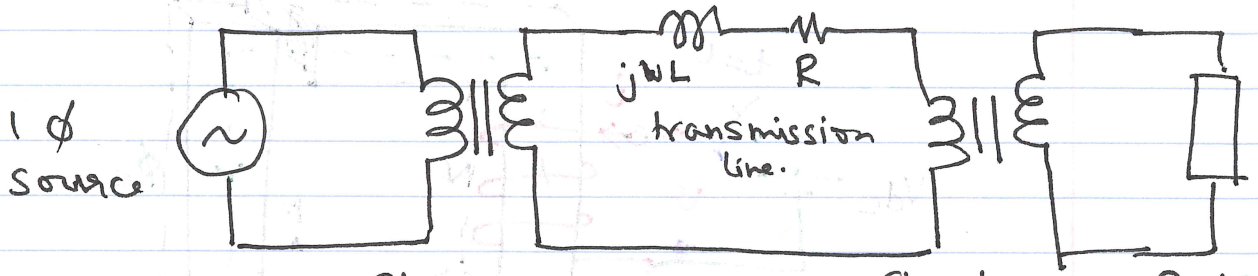
Lecture 16.

Last time: Transformers
- Coupled circuit
- Equivalent circuit.

Q. Why did we find out the electric equivalent circuit?



Step up transformer ⇓ Substitute with eqv. ckt. ⇓ phasors

Step down transform ⇓ substitute with eqv. ckt. ⇓ phasors

P+jQ load.

Solve power flow, $\bar{V}$, $\bar{I}$.

$\frac{d}{dt} = j\omega$.

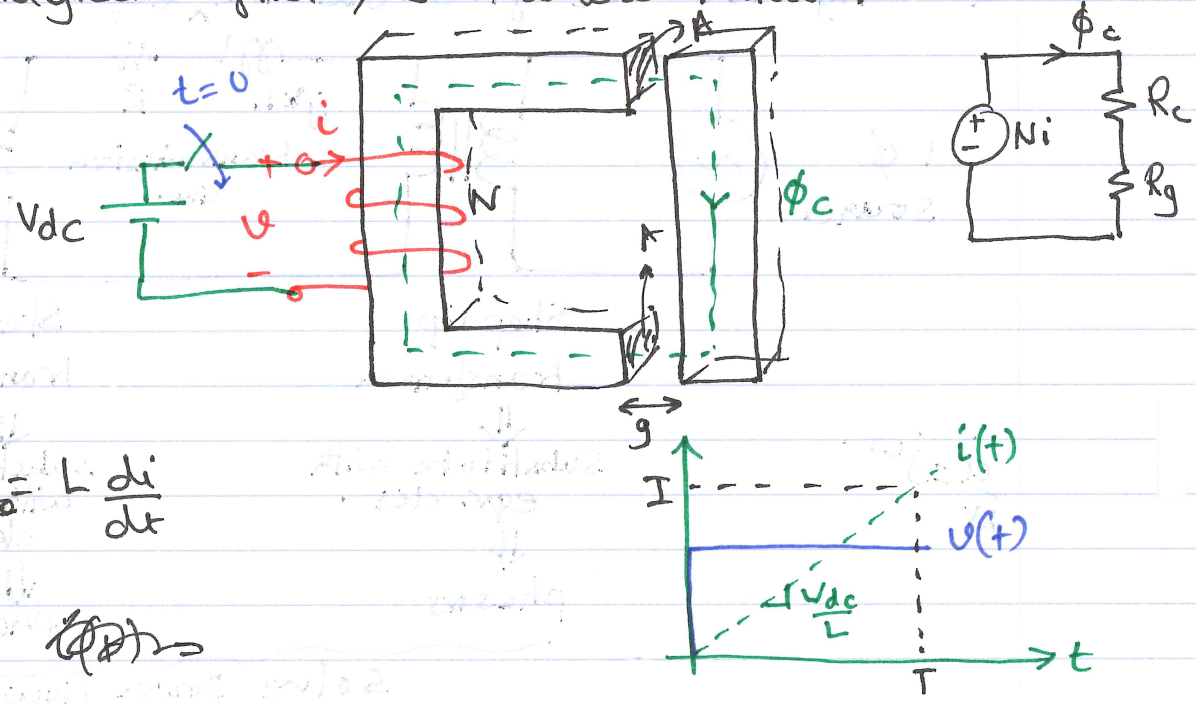
Q. Can we use transformers in dc transmission system? Think!!

Electromechanical Energy Conversion:

Applications: ① Motors, generators, breakers, speakers. Nearly half the electrical energy we produce goes into a motor for industrial, commercial, residential application

Before we go try to understand, how we can convert electricity to mechanical motion and vice versa,

Q. We know an inductor stores energy in the magnetic field, but where? how?



$$v(t) = L\frac{di}{dt}$$

Instantaneous electrical power, $p(t) = v(t) \cdot i(t)$.

∴ Energy store over a period of T.

$$\mathcal{E} = \int_0^T p(t)\,dt = \int_0^T v(t)\cdot i(t)\,dt$$

$$= \frac{1}{2} L I^2$$

$$= \frac{1}{2}\frac{N^2 I^2}{(R_c+R_g)} = \frac{1}{2}\phi_c^2 \cdot (R_c+R_g)$$

$$= \frac{1}{2}\phi_c^2\left[\frac{l_c}{\mu_0\mu_r A_c} + \frac{g}{\mu_0 A_g}\right]$$

$R_c = \frac{l_c}{\mu_0\mu_r A_c}$

$R_g = \frac{l_g}{\mu_0 A_g}$

$A_g = A_c = A$

$$= \frac{1}{2} \frac{\phi_c^2}{A^2} \left[\frac{l_c A}{\mu_0 \mu_r} + \frac{g \cdot A}{\mu_0} \right]$$

$$= \frac{1}{2} B^2 \left[\frac{V_c}{\mu_0 \mu_r} + \frac{V_g}{\mu_0} \right]$$

V_c → Volume of the core

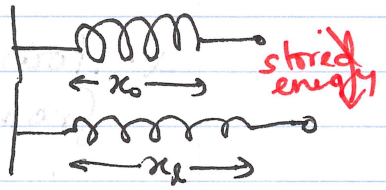
V_g → Volume of the gap.

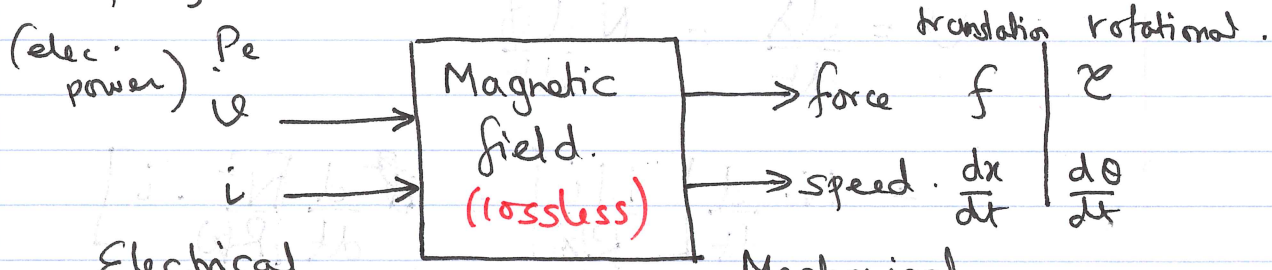
As $\mu_r \gg 1$.

Energy stored in core $\ll$ Energy stored in the gap.

Q. Can we use this energy stored in the magnetic field to do something?

For example, imagine a spring. Now if we store energy in the spring, it can do work.

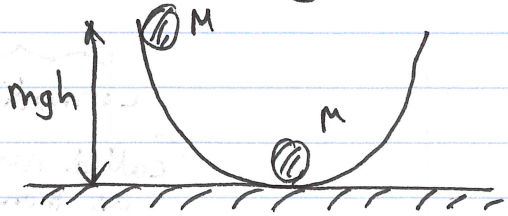


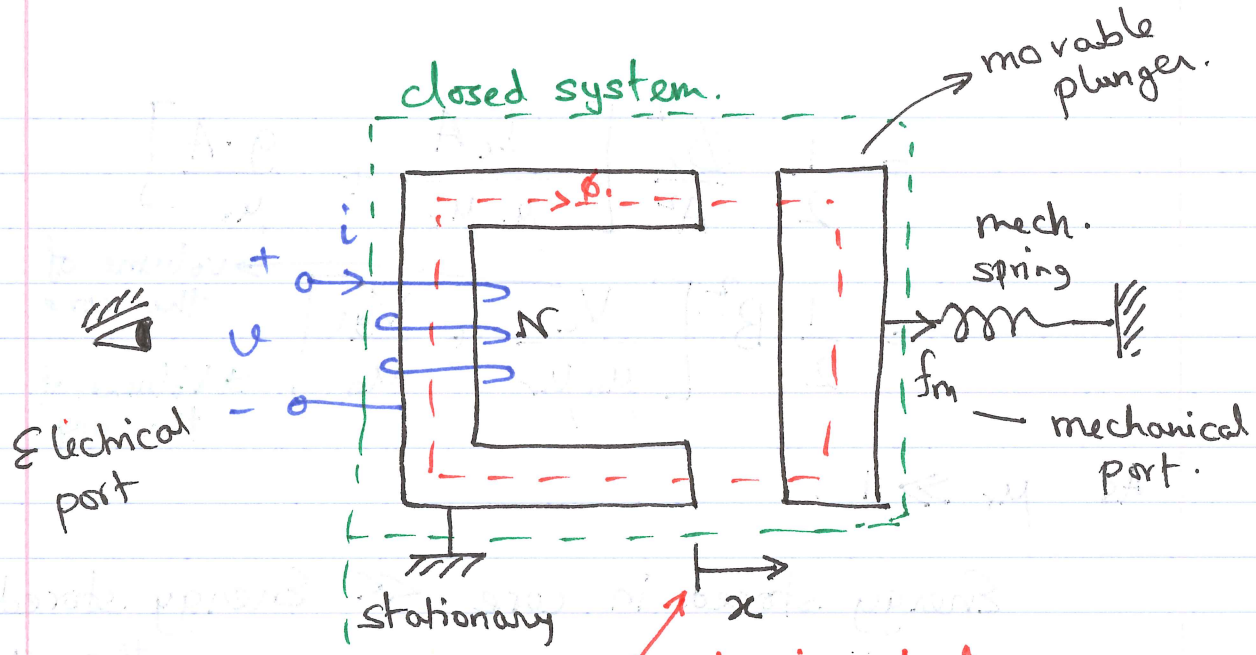


Two-port network

Q. What will be the natural tendency of the closed system?

Minimize stored energy. How? close the gap.





Q. How does this closed system look like from the electrical port perspective?

$$v = \frac{d\lambda}{dt} = N\frac{d\phi}{dt}$$

$$= N \cdot \frac{d}{dt}\left[\frac{Ni}{R(x)}\right] = \frac{d}{dt}\left[\frac{N^2}{R(x)} \cdot i\right]$$

$$= \frac{d}{dt}\left[L(x) \cdot i\right]$$

velocity.

$$= L(x)\frac{di}{dt} + i\frac{d}{dx}L(x)\cdot\frac{dx}{dt}$$

conventional. (called here as transformer voltage)

speed voltage.

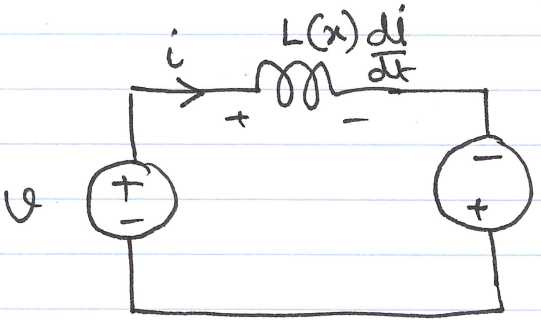
Assume core infinitely permeable.

Q. What is

$$L(x) = \frac{N^2}{R_g} = \frac{N^2}{2x}\mu_0 A .$$

$$\therefore \quad v = \mu_0 \frac{N^2 A}{2x}\frac{di}{dt} - i\,N^2\mu_0 \frac{A}{2x^2}\frac{dx}{dt}$$

→ sign — super important



E_b ← back emf

$E_b = N^2 i \mu_0 A \cdot \frac{dx}{dt} \cdot \frac{1}{2x^2}$

Physically: If $\frac{dx}{dt} > 0$, the plunger is moving away., reluctance increases, flux decreases for same Ni, so the back emf must aid the voltage source, to draw more current, such that flux can be maintained (Lenz's Law).

